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LINCOLN, NEBRASKA, June 1, 1898.

To the Hon. Silas A. Holcomb, Governor of Nebraska :

SIR—In accordance with the provisions of law, we herewith submit our report of the proceedings of the State Historical Society for the past year.

Very respectfully,

J. STERLING MORTON,

President.

HOWARD W. CALDWELL,

Secretary.

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THE BARITES OF NEBRASKA AND THE BAD LANDS.

ERWIN HINCKLEY BARBOUR.

During the past two years there have been several additions to the list of Nebraska minerals. Chief among them are the closely related minerals Barite and Celestite. Because of the larger collections and the better knowledge of the barites they will be made the subject of this paper.

Our barites occur in three rather distinct forms, the flat or tabular crystals of southeastern Nebraska; the superb prismatic crystals of the "Bad Lands," and the fibrous dike barite found in the Hat Creek basin of the Little Bad Lands in Sioux county. In southeastern Nebraska, in the Permian of Gage county, near Odell, Wymore, and Beatrice, barites are found quite abundantly in the clays. Because of their shape and frequent occurrence these beds have become known locally as the "Diamond Fields."

The barite group belongs crystographically to the orthorhombic system, that is, the three axes are all at right angles, but are each of different length, accordingly the prism may be developed along different axes, making ever varying forms of crystals. The common form of crystal is flat and tabular, sometimes, however, they are long and needle like, at other times thick, strong prisms. The form found in Gage county is a flat, diamond-shaped crystal from one to three millimeters in thickness. The largest of these will scarcely measure more than ten to fifteen millimeters (three-eighths to one-half inch) in length. It is a common feature of these crystals to show alternating bands of white, brown, or yellow color, parallel to the edges of the crystal; also a dark cross imitating axes is often present. The yellow portion, according to Dana, is the less pure barite, being, in fact, a pretty nearly equal mixture of barium sulphate and calcium carbonate. In addition to these it is not uncommon for barite to contain impurities in the way of silica, clay, and bituminous or carbonaceous substances. The more transparent crystals show phantom figures to perfection.

These are all the more interesting to the Nebraska mineralogist from the fact that they are undescribed for the state.

In the Bad Lands there are thin dikes running in all directions, over the hills. These are generally dikes of chalcedony, and stand but little above the bare clays of the region. In other cases there is a filling of calcite with selvages of chalcedony. There are besides, occasionally dikes of sandstone a half meter thick, and even dikes of clay.

On the last expedition sent out by the State University,—the Morrill Geological Expedition of 1895,—the students found uncommon and altogether unexpected dikes of fibrous barite of a bluish color. The dike was scarcely more than fifteen to twenty millimeters (a half to three-quarters of an inch) thick, yet it could be traced for some distance across the Bad Land marls. The dike was vertical, the fibers at right angles to its plane. This is the first known occurrence of Barite in any form in this locality.

Further north in the Big Bad Lands a magnificent array of barites is found in the Fort Pierre shale. These are occasionally of striking size and of great superiority of color and crystallization. The prevailing type is a long, tapering, prismatic crystal of a fine amber color. There are occasional crystals found which are almost pure and transparent. Some are less distinctly crystallized and are arranged in conspicuously radiated bunches. The mode of occurrence is an interesting feature. Wherever the country is cut into hills by recent drainage lines, one can trace along the hillsides a band made conspicuous by its nodules or concretions. These vary in size from the diameter of the fist to those exceeding that of the outstretched arms. These are exposed along the Cheyenne river and its many tributaries. The concretions are of that peculiar type known as septaria. There is a well recognized tendency of matter in solution,—hence free to arrange itself molecularly,—to become segregated, or aggregated together around a center, making more or less spherical masses. Thus it is that we find in the shales of the Fort Pierre Cretaceous great concretions of the same material, though rendered hard and dense. These clay balls when drying from the original plasticity, harden first on the surface. Naturally then, as the interior dried there would be all sorts of shrinkage cracks and irregular cavities left within. Here we

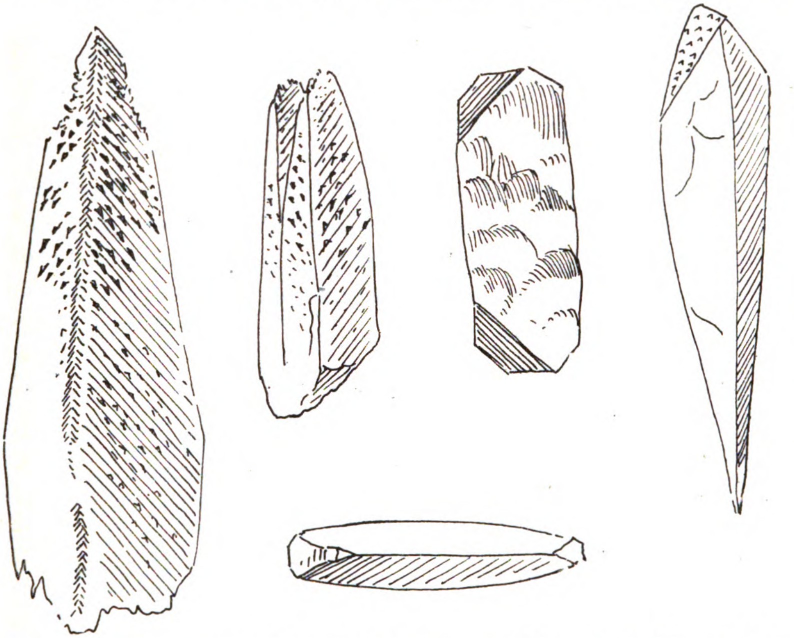

FIGURE 2.—A group of amber-colored barite crystals from the Bad Lands, showing etched and doubly terminated forms. The natural etching of these crystals is often rather remarkable. Doubly terminated crystals are rare. Natural size.

have formed a beautiful receptacle for the magnificent crystals which are to be formed within these drying mud-balls. Soon water with calcium carbonate in solution coats all the surfaces with a layer of impure and discolored calcite. Succeeding layers are of better color and crystallization. Generally the cavities are lined with small crystals of dog-tooth spar of an orange color; upon these rest clear, sharp crystals of nail-head calcite, and also the fine tapering barite crystals. In breaking open these flinty clay balls it is a difficult matter to avoid jarring and breaking the slender crystals within. There are hundreds of these concretions in sight, although but few contain the barite, and the matter of collecting barites is reduced to faithfulness in opening numerous concretions. Sometimes these septaria are so hollow and bristling with crystals that the whole is very geode-like.

Out of many hundred crystals but few doubly terminated ones were secured. The crystals are often etched in a remarkably clean-cut and beautiful manner, the etchings all pointing in a given direction and with definite and unvarying relation to the axes.

It was the author's good fortune to have visited this region before it became known to collectors, and in this way he secured first choice of these beautiful crystals. A more technical study of our western Barites has been begun and will be ready for publication at another time.

December 18, 1896.

PLATE II, Figs. 1 to 6.—A group of barite crystals from the Bad Lands, sketched natural size. The radiated form shown in Fig. 5 is occasionally met. Fig. 6, the form of crystal found in the white and transparent barite of the region, which is rather rare. The others are of the amber-colored type. The superficial characters are apparent without descriptions.

PLATE III.—A group of barite crystals from the "Diamond Fields" of Gage county, Nebraska, magnified about three diameters. All viewed by reflected light.

PLATE IV.—A group of barite crystals from Gage county, Nebraska, magnified about three diameters. All viewed by transmitted light.

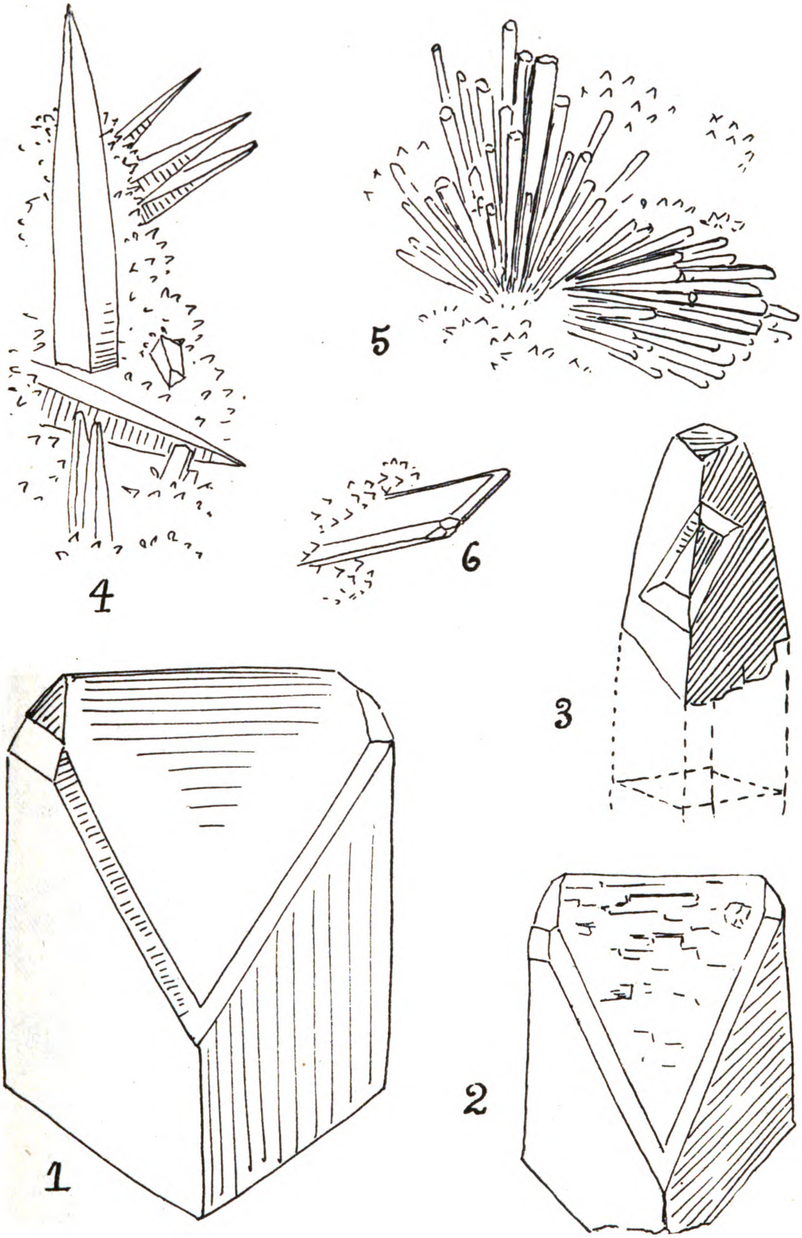

PLATE II.

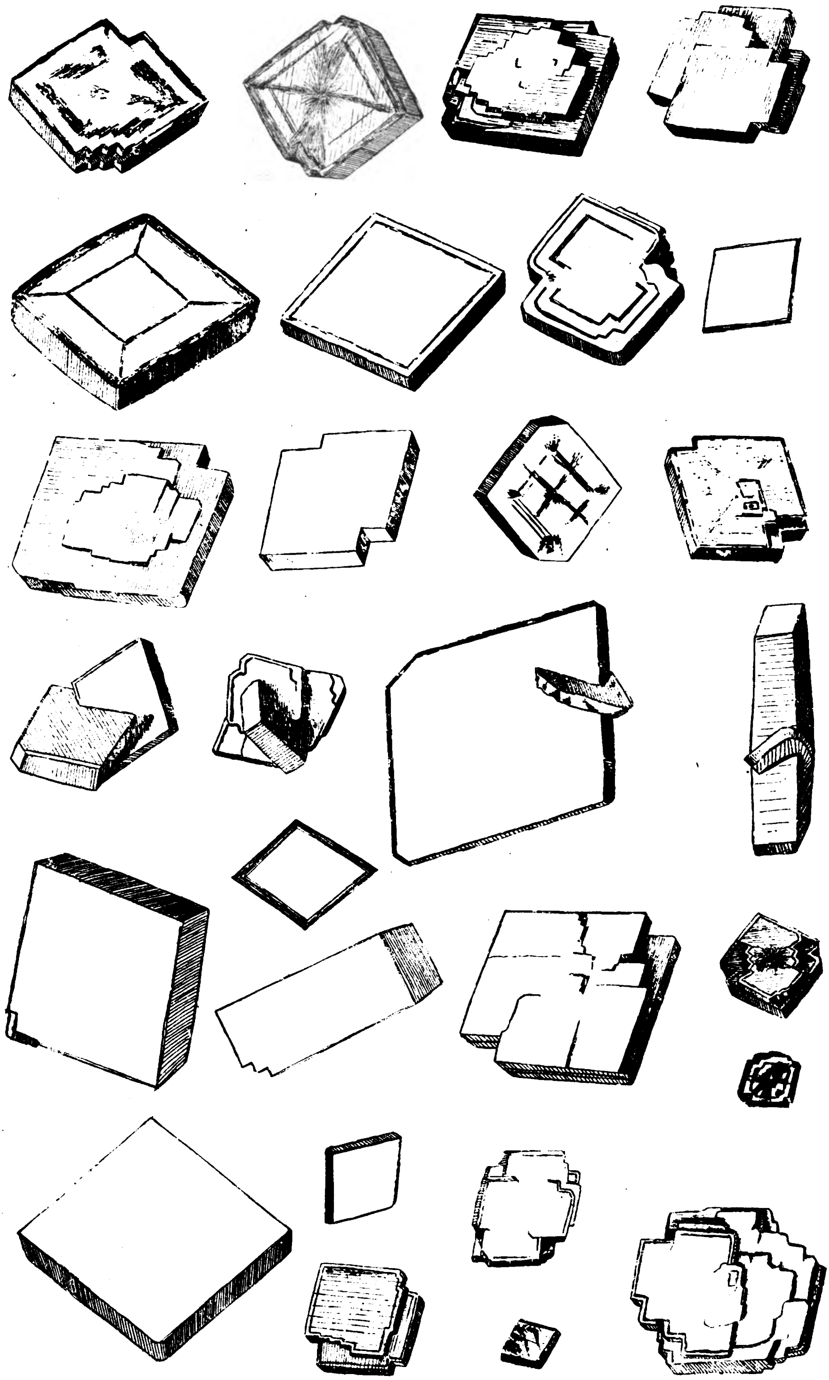

PLATE III.

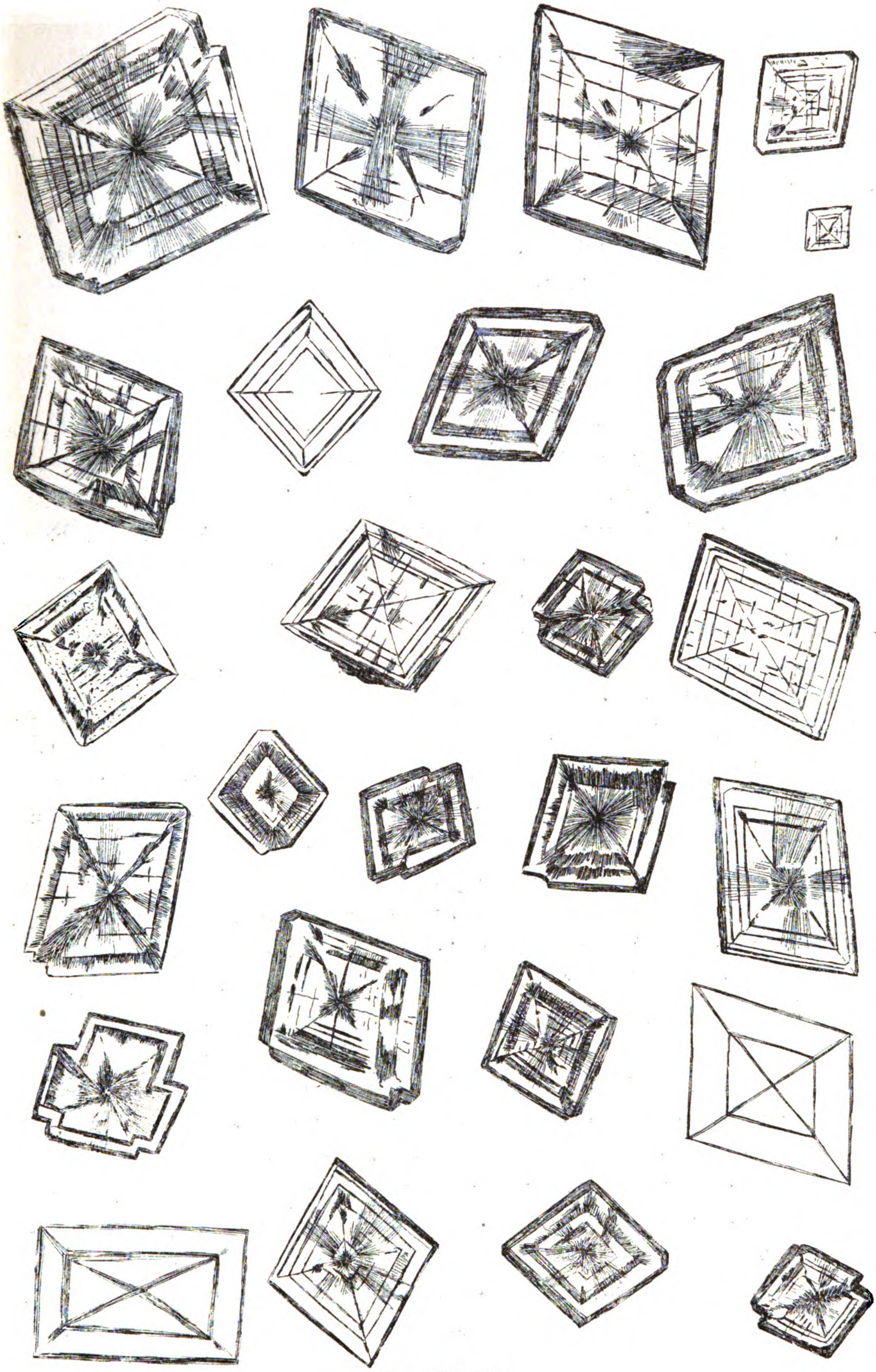

PLATE IV.